



OncoBind

Every kinase inhibitor binds the same target differently. We show physicians exactly how — and which one fits their patient.

🕒 THE MOMENT

🕒 TUESDAY, 2:40PM — ONCOLOGY CLINIC

A patient just tested positive for an **EGFR exon 19 deletion**. Four FDA-approved drugs target it. The physician has treated maybe a dozen cases like this — not enough to have felt the real-world difference between them. Each drug's trial data lives in a different paper. None of them show her **how the drugs actually bind this patient's specific mutation**, side by side.

She has to choose today. The tools available to help her haven't changed in a decade.

This isn't one physician's problem. It's structural.

- Molecular subtyping now routinely identifies the exact mutated target — but for most such targets there are now **three to five FDA-approved drugs**, not one, and the count keeps growing.
- Selection is guided largely by approval recency and clinician familiarity, **not comparative structural or binding data**, which is scattered across separate primary-literature sources for each drug.
- Adverse-effect tradeoffs — a black-box cardiovascular warning versus a milder but less potent alternative — are buried in **separate, non-comparative prescribing documents**, not surfaced at the moment of decision.

5 major kinase-target families. 17+ approved drugs across them. No structure-based comparison tool built for the point of care — until now.

The models to answer this got fast enough to matter, this year.



01

Models got fast

AlphaFold-era docking and affinity models (DiffDock, Boltz2, and peers) now return a predicted binding pose and affinity in minutes — down from weeks of wet-lab crystallography.



02

Models became API-accessible

Hosted inference platforms (NVIDIA BioNeMo among others) put these models behind a simple API — no in-house computational chemistry team or GPU cluster required.



03

The drug landscape exploded

More kinase-inhibitor approvals in the last five years than the prior fifteen — each one adding to the comparison problem, none of them making it easier to solve.

OncoBind turns that decision into a structural comparison, live.

For a diagnosed molecular subtype, OncoBind docks every approved candidate drug against the real target structure, predicts binding affinity, and combines it — transparently — with adverse-effect burden and the individual patient's context into one explainable score.



Dock

DiffDock predicts the real 3D binding pose of each candidate against the real receptor structure.



Predict

Boltz2 predicts binding affinity for the same ligand-target pair — same model class that redefined structural biology.



Score

A transparent, published formula combines affinity, docking confidence, and adverse-effect burden.



Personalize

Patient-specific factors re-weight the score live — the same drug ranks differently for different patients.

Five real scenarios today — any drug, live, on demand


OncoBind
 Precision Oncology Therapy Advisor

built on  Base44

MOLECULAR TARGET · THERAPY COMPARISON

Choose the therapy the biology actually supports.

For each molecularly-defined cancer, OncoBind docks every approved candidate drug against the real target structure, predicts binding affinity, and weighs the result against known adverse-effect burden — so the tradeoffs are visible, not buried in prescribing information.

🔗 **How to read this:** the **composite score** (0–100) blends predicted binding affinity, docking confidence, and adverse-effect burden — higher is a stronger candidate *relative to the other drugs in that scenario only*. The 3D viewer shows the real target structure fetched live from RCSB PDB; once a drug's binding pose is computed, it overlays as a docked ligand.

⚠️ **Research & education tool only.** Binding poses and affinities are computational predictions, not clinical measurements. This is not a diagnostic or prescribing device and does not replace clinical judgment, current FDA prescribing information, or institutional guidelines.


NON-SMALL CELL LUNG CANCER (NSCLC)


EGFR-Mutant Non-Small Cell Lung Cancer
 EGFR exon 19 deletion / L858R activating mutations
 Target: EGFR Tyrosine Kinase Domain · PDB 1M17


CUTANEOUS MELANOMA


BRAF V600E-Mutant Melanoma
 BRAF V600E activating mutation
 Target: BRAF V600E Kinase Domain · PDB 3OG7


BREAST CANCER


HER2-Positive Breast Cancer
 ERBB2 (HER2) gene amplification/overexpression
 Target: HER2 (ERBB2) Tyrosine Kinase Domain · PDB 3PP0


CHRONIC MYELOID LEUKEMIA (CML)


Chronic Myeloid Leukemia (BCR-ABL1+)
 BCR-ABL1 fusion oncoprotein (Philadelphia chromosome)
 Target: ABL1 Tyrosine Kinase Domain · PDB 2HYY

EGFR-mutant NSCLC, BRAF V600E melanoma, HER2+ breast cancer, CML, and ALK+ NSCLC — each backed by a real crystallized target structure. **Type in any drug not on the list and OncoBind docks it live** against the same target, in real time.

Same drug list. Different patient. Different answer.

RELEVANT PATIENT FACTORS

☒ Cardiac history / QT risk
 ☐ Hepatic impairment

☐ Renal impairment
 ☐ CNS involvement / brain metastases

☐ Prior TKI exposure / resistance concern

OR UPLOAD A LAB REPORT (OPTIONAL)

DRUG	CLASS	DOCKING CONF.	PIC50	PATIENT-ADJUSTED SCORE	KEY ADVERSE EFFECTS
Ponatinib Iclusig	3rd-generation pan-BCR-ABL1 TKI	0.32	12.15	49 (-26)	Arterial occlusive events Hepatotoxicity
Dasatinib Sprycel	2nd-generation dual BCR-ABL1/SRC TKI	-0.26	12.06	45 (-8)	Pleural effusion Thrombocytopenia
Imatinib Gleevec	1st-generation BCR-ABL1 TKI	0.13	9.64	40	Periorbital edema Nausea
Nilotinib Tasigna	2nd-generation BCR-ABL1 TKI	0.42	10.14	12 (-32)	QT prolongation Hyperglycemia

Flag a patient's cardiac history and watch **Nilotinib fall from 44 to 12** — its QT-prolongation black-box warning now weighs directly on the score, with the exact match shown. Every adjustment traces to real, cited data. No hidden model judgment.

No black box. Every number traces to a named model or a public database.

Unlike opaque "AI risk score" tools, OncoBind's scoring is a published, auditable formula — the same one its in-app AI advisor is instructed to explain verbatim, on request, to any physician who asks.

```
// composite_score, normalized within each scenario's candidate drugs
composite_score = 100 × clamp(0.45 × affinity_norm + 0.35 × docking_norm + 0.20 × (1 - adverse_norm))
// patient_adjusted_score layers on top, per flagged clinical factor
patient_adjusted = clamp(composite_score - matched_risk_penalty + matched_relevance_bonus, 0, 100)
```

Every molecular tumor board is a potential user — three paths to revenue.

17+

approved drugs already mapped across just 5 target families in the live product

Accelerating

more kinase-inhibitor approvals in the last 5 years than the prior 15 — the market grows with the pipeline it depends on

3

revenue paths, sequenced by validation risk, sharing one engine

NEAR-TERM

Institutional SaaS

Per-site or per-seat subscription for health systems and oncology practices with molecular tumor boards.

MID-TERM

Data & API licensing

Structured binding/affinity comparison data licensed to pharma medical-affairs and payer/formulary teams.

LONG-TERM

EHR-embedded CDS

Integration into oncology EHR workflows as point-of-care clinical decision support.

PLACEHOLDER — NEEDS SOURCED DATA

Quantified TAM/SAM/SOM and pricing to be added from a licensed market-research source before external distribution of this deck.

Structural, comparative, and transparent — a gap the incumbents don't fill

CATEGORY	STRENGTH	GAP ONCOBIND FILLS
Static reference tools (NCCN, Lexicomp, UpToDate)	Comprehensive, trusted, guideline-anchored	Not structural or comparative — read one drug at a time, no binding-based ranking, no patient-specific adjustment
OncoBind	Structure-based, transparent, explainable, personalized per-drug scoring	—
Black-box AI risk tools	Data-driven, can flag risk patterns	Opaque scoring, heavy regulatory burden, no grounding in physical binding data

From 5 scenarios to a platform

NEAR-TERM

Expand coverage + pilot

Grow beyond the initial 5 scenarios / 17 drugs to the full set of clinically relevant targeted-therapy classes; validate the workflow with a design-partner health system.

LONG-TERM

Broader pharmacogenomic CDS

Extend beyond oncology-target binding into patient-specific pharmacogenomic interaction data and EHR-embedded workflows.

PLACEHOLDER

Founding team & current traction

Add founder bios and pilot/LOI status here.

PLACEHOLDER

The ask & use of funds

Add raise amount, instrument, and allocation here.

This deck describes a working technical prototype — every screenshot is the real, live product. Sections marked "Placeholder" intentionally contain no fabricated figures; fill in with real, current data before sharing externally.



**Dr. Chen shouldn't have to choose blind.
Neither should the next physician, or the
one after that.**

oncobind.app · [founder email placeholder]